## A phase 2b, randomized, placebo-controlled, double-blind, dose-ranging study of the neurokinin 3 receptor antagonist fezolinetant for vasomotor symptoms associated with menopause.

*PMID: 32102086 | Indexed in PubMed*

Menopausal vasomotor symptoms (VMS) may result from altered thermoregulatory control in brain regions innervated by neurokinin 3 receptor-expressing neurons. This phase 2b study evaluated seven dosing regimens of fezolinetant, a selective neurokinin 3 receptor antagonist, as a nonhormone approach for the treatment of VMS. Menopausal women aged >40-65 years with moderate/severe VMS (≥50 episodes/wk) were randomized (double-blind) to fezolinetant 15, 30, 60, 90 mg BID or 30, 60, 120 mg QD, or placebo for 12 weeks. Primary outcomes were reduction in moderate/severe VMS frequency and severity ([number of moderate VMS × 2] + [number of severe VMS × 3]/total daily moderate/severe VMS) at weeks 4 and 12. Response (≥50% reduction in moderate/severe VMS frequency) was a key secondary outcome. Of 352 treated participants, 287 completed the study. Fezolinetant reduced moderate/severe VMS frequency by -1.9 to -3.5/day at week 4 and -1.8 to -2.6/day at week 12 (all $P < 0.05$ vs placebo). Mean difference from placebo in VMS severity score was -0.4 to -1 at week 4 (all doses $P < 0.05$) and -0.2 to -0.6 at week 12 ($P < 0.05$ for 60 and 90 mg BID and 60 mg QD). Response (50% reduction) relative to placebo was achieved by 81.4% to 94.7% versus 58.5% of participants at end of treatment (all doses $P < 0.05$). Treatment-emergent adverse events were largely mild/moderate; no serious treatment-related treatment-emergent adverse events occurred. Fezolinetant is a well-tolerated, effective nonhormone therapy that rapidly reduces moderate/severe menopausal VMS. : Video Summary:http://links.lww.com/MENO/A572; video script available at http://links.lww.com/MENO/A573.

## Fezolinetant: A Potential Treatment for Moderate to Severe Vasomotor Symptoms of Menopause.

*PMID: 38187079 | Indexed in PubMed*

The most common symptom of menopause is vasomotor symptoms (VMS), which occur in more than 80% of postmenopausal women. Furthermore, VMS are the manifestation of menopause for which women most commonly seek treatment, namely, to address their impacted quality of life, including sleep, and work-and non-work-related productivity. VMS vary in frequency, intensity and duration. Hormone therapy (HT) has been our most effective treatment for VMS and has been approved for this indication by the United States Food and Drug Administration (FDA). Despite being a safe and effective treatment option, many patients and providers are hesitant to consider HT. Moreover, HT is contraindicated for some women. While many over-the-counter and non-HT options are available, we lack data on the efficacy and safety of most of these. This has left a void for women. Fezolinetant was recently approved by the FDA for the treatment of moderate-to-severe VMS. So far, clinical trials have shown positive results in terms of safety and efficacy. Fezolinetant is a non-hormonal, neurokinin 3 receptor antagonist that works in the hypothalamus at the thermoregulatory centre. Blocking the non-hormonal neurokinin 3 receptor antagonist modulates hot flashes and night sweats. As early as 4 weeks from initiating fezolinetant, women experienced a statistically significant reduction of both severity and frequency of VMS per day, resulting in an improved quality of life.

## FDA approved fezolinetant (Veozah): a critical evaluation of its efficacy and safety for menopausal vasomotor symptoms, calling for prospective research.

*PMID: 38478035 | Indexed in PubMed*

Women going through menopause frequently experience vasomotor symptoms such as hot flashes, night sweats, and sleep disturbances, significantly influencing their quality of life. Hormonal therapy has been demonstrated to be beneficial in treating VMS. However, due to specific restrictions, it is not recommended for every woman. Fezolinetant, a neurokinin 3 antagonist and non-hormonal treatment for severe to moderate VMS, functions by inhibiting neuronal impulses originating from the hypothalamic thermoregulatory center. Current Skylight 2 and 4 trials statistically demonstrate the safety and acceptability of fezolinetant, with relatively few adverse effects reported. Fezolinetant has been shown great potential for treating menopausal-related VMS, supporting its further advancement. However, further investigation is required to thoroughly evaluate its safety, effectiveness, and its impact on sleep patterns.

## Medium-chain triglyceride supplementation and the association with growth, nutritional status and clinical outcomes in infants with biliary atresia.

*PMID: 39667195 | Indexed in PubMed*

Infants with biliary atresia experience gastrointestinal malabsorption of long-chain triglycerides and are commonly supplemented with medium-chain triglyceride (MCTs) that can be passively absorbed. The aim was to investigate the association of MCT supplementation with growth, nutritional status and clinical outcomes in infants with biliary atresia. Infants who underwent Kasai portoenterostomy and were followed up for at least two years or until death or transplantation were reviewed. Infants with comorbidities affecting growth or outcome were excluded. Data were extracted from medical records from more than a decade in relation to MCT supplementation, growth, nutritional status and clinical outcome at baseline, 6-weeks, 3-, 6-, 12- and 24-months. Mixed-effects modelling was used to test associations of MCT in the first six months with these outcomes. Of 200 infants (108 male), 108 (54 %) were alive with native liver at two years, 84 (42 %) underwent liver transplantation and eight (4 %) died. MCT percentage prescribed was mean 57.3 % (SD 11.2) while MCT intake was median 2.7 (IQR 2.2, 3.8) g/kg/d. For every g/kg/d MCT consumed, the rate of change in z-score for weight was -0.27 (95 % CI -0.37 to -0.17) and length was -0.31 (-0.42 to -0.17) (both $p < 0.001$). Compared to the low MCT group (<2.7 g/kg/d), the high group (≥2.7 g/kg/d) consumed more energy (118 vs. 108 kcal/kg; $p < 0.001$), however, at 3-months they had lower weight (-1.7 (1.2) v. -1.0 (1.2) and length (-1.3 (1.1) v. -0.6 (1.4) z-scores (both $p < 0.001$) but no differences in growth at later time points. There was no overall association between MCT and nutritional status or clinical outcomes. This is the first study to investigate the association of MCT with growth, nutritional status and clinical outcomes in biliary atresia. No association was found between MCT with growth beyond 3-months, overall nutritional status or clinical outcomes. The association between MCT (g/kg/d) and poorer growth in the first 3-months may be explained by infants with poorer growth drinking more or being prescribed more MCT formula milk. A randomised controlled trial could help to better understand this association.

## Microcystin-RR is a biliary toxin selective for neonatal extrahepatic cholangiocytes.

*PMID: 39687604 | Indexed in PubMed*

Biliary atresia is a fibrosing cholangiopathy affecting neonates that is thought to result from a prenatal environmental insult to the bile duct. Biliatresone, a plant toxin with an α-methylene ketone group, was previously implicated in biliary atresia in Australian livestock, but is found in a limited location and is unlikely to be a significant human toxin. We hypothesized that other unsaturated carbonyl compounds, some with the potential for significant human exposure, might also be biliary toxins. We focused on the family of microcystins, cyclic peptide toxins from blue-green algae that are found worldwide, particularly during harmful algal blooms. We used primary extrahepatic cholangiocyte spheroids and extrahepatic bile duct explants from both neonatal [a total of 86 postnatal day (P) 2 mouse pups and 18 P2 rat pups (n = 8-10 per condition for both species)] and adult rodents [a total of 31 P15-18 mice (n = 10 or 11 per condition)] to study the biliary toxicity of microcystins and potential mechanisms involved. Results showed that 400 nM microcystin (MC)-RR, but not six other microcystins or the related algal toxin nodularin, caused >80% lumen closure in cell spheroids made from extrahepatic cholangiocytes isolated from 2-3-day-old mice ($p < 0.0001$). By contrast, 400 nM MC-RR resulted in less than an average 5% lumen closure in spheroids derived from neonatal intrahepatic cholangiocytes or cells from adult mice ($p = 0.4366$). In addition, MC-RR caused occlusion of extrahepatic bile duct explants from 2-day-old mice ($p < 0.0001$), but not 18-day-old mice. MC-RR also caused a 2.3-times increase in reactive oxygen species in neonatal cholangiocytes ($p < 0.0001$), and treatment with N-acetyl cysteine partially prevented microcystin-RR-induced lumen closure ($p = 0.0004$), suggesting a role for redox homeostasis in its mechanism of action. We identified MC-RR as a selective neonatal extrahepatic cholangiocyte toxin and suggest that it acts by increasing redox stress. The plant toxin biliatresone causes a biliary atresia-like disease in livestock and vertebrate animal model systems. We tested the widespread blue-green algal toxin, microcystin-RR, another highly electrophilic unsaturated carbonyl compound that is released during harmful algal blooms, and found that it was also a biliary toxin with specificity for neonatal extrahepatic cholangiocytes. This work should drive further animal studies and, ultimately, studies to determine whether human exposure to microcystin-RR causes biliary atresia.

## Biliary atresia with splenic malformation with associated ductal plate malformation and duodenal atresia: A case report.

*PMID: 38394407 | Indexed in PubMed*

Biliary atresia (BA) is the most common cause of the obstructive type of neonatal cholestasis that requires prompt surgical intervention. About 10% of neonates with BA have other congenital anomalies, of which splenic malformation (BASM) is a well-known distinct sub-group. There is sparse literature on the association of duodenal atresia and ductal plate malformation (DPM) in patients of BASM. We describe a BASM associated with DPM and duodenal atresia in a 35-day-old infant, who succumbed at 40 days, before portoenterostomy could be performed. Duodenal atresia can be one of the associated malformations associated with BASM and ductal plate abnormalities. In our case, the child did not survive.

## Progress in Biomarkers Related to Biliary Atresia.

*PMID: 38426193 | Indexed in PubMed*

Biliary atresia (BA) is a congenital cholestatic disease that can seriously damage children's liver function. It is one of the main reasons for liver transplantation in children. Early diagnosis of BA is crucial to the prognosis of patients, but there is still a lack of reliable non-invasive diagnostic methods. Additionally, as some children are in urgent need of liver transplantation, evaluating the stage of liver fibrosis and postoperative native liver survival in children with BA using a straightforward, efficient, and less traumatic method is a major focus of doctors. In recent years, an increasing number of BA-related biomarkers have been identified and have shown great potential in the following three aspects of clinical practice: diagnosis, evaluation of the stage of liver fibrosis, and prediction of native liver survival. This review focuses on the pathophysiological function and clinical application of three novel BA-related biomarkers, namely MMP-7, FGF-19, and M2BPGi. Furthermore, progress in well-known biomarkers of BA such as gamma-glutamyltransferase, circulating cytokines, and other potential biomarkers is discussed, aiming to provide a reference for clinical practice.

## Aetiology of biliary atresia: what is actually known?

*PMID: 23987231 | Indexed in PubMed*

Biliary atresia (BA) is a rare disease of unknown etiology and unpredictable outcome, even when there has been timely diagnosis and exemplary surgery. It has been the commonest indication for liver transplantation during childhood for the past 20 years. Hence much clinical and basic research has been directed at elucidating the origin and pathology of BA. This review summarizes the current clinical variations of BA in humans, its occasional appearance in animals and its various manifestations in the laboratory as an experimental model.

## Biliary atresia.

*PMID: 19914515 | Indexed in PubMed*

Biliary atresia is a rare disease of infancy, which has changed within 30 years from being fatal to being a disorder for which effective palliative surgery or curative liver transplantation, or both, are available. Good outcomes for infants depend on early referral and timely Kasai portoenterostomy, and thus a high index of suspicion is needed for investigation of infants with persistent jaundice. In centres with much experience of treating this disorder, up to 60% of children will achieve biliary drainage after Kasai portoenterostomy and will have serum bilirubin within the normal range within 6 months. 80% of children who attain satisfactory biliary drainage will reach adolescence with a good quality of life without undergoing liver transplantation. Although much is known about management of biliary atresia, many aspects are poorly understood, including its pathogenesis. Several hypotheses exist, implicating genetic predisposition and dysregulation of immunity, but the cause is probably multifactorial, with obliterative extrahepatic cholangiopathy as the common endpoint. Researchers are focused on identification of relevant genetic and immune factors and understanding serum and hepatic factors that drive liver fibrosis after Kasai portoenterostomy. These factors might become therapeutic targets to halt the inevitable development of cirrhosis and need for liver transplantation.

## Biliary atresia with rare associations: a case report and literature review.

*PMID: 39525756 | Indexed in PubMed*

Biliary atresia is a rare, progressive cholangiopathy that affects newborns, causing jaundice and other manifestations of hyperbilirubinemia. The incidence is higher in Asia than in Europe. The only available treatment is a surgical operation called Kasai portoenterostomy. In this case, the authors highlighted rare congenital anomalies that came with biliary atresia. A 10-day-old male infant was admitted to the hospital due to recurrent vomiting, yellowish skin, and scleral icterus. Laboratory investigations revealed elevated total serum and direct bilirubin levels. An atrophic gallbladder was observed on ultrasound. Intrahepatic cholangiography confirmed the diagnosis of biliary atresia, leading to the performance of a Kasai procedure. Additionally, the patient had intestinal malrotation and volvulus, which were managed with a Ladd's procedure. Following surgery, there was notable improvement in liver enzymes and bilirubin levels, and the patient was discharged after 7 days. The infant has been initiated on oral vitamins, ursodeoxycholic acid, and antibiotics. Biliary atresia is a challenging condition characterized by progressive narrowing and fibrosis of the biliary tree. It is rarely associated with rare congenital anomalies like situs inversus totalis, intestinal malrotation, and volvulus. Diagnosis involves abdominal ultrasound and MRCG. The biliary atresia was managed by the Kasai procedure and the intestinal malrotation, and volvulus were managed by Ladd's procedure. This case report highlights the importance of considering rare associations such as situs inversus, intestinal malrotation and volvulus in the diagnosis of biliary atresia in newborn. Early diagnosis and prompt intervention are crucial for optimal outcomes.